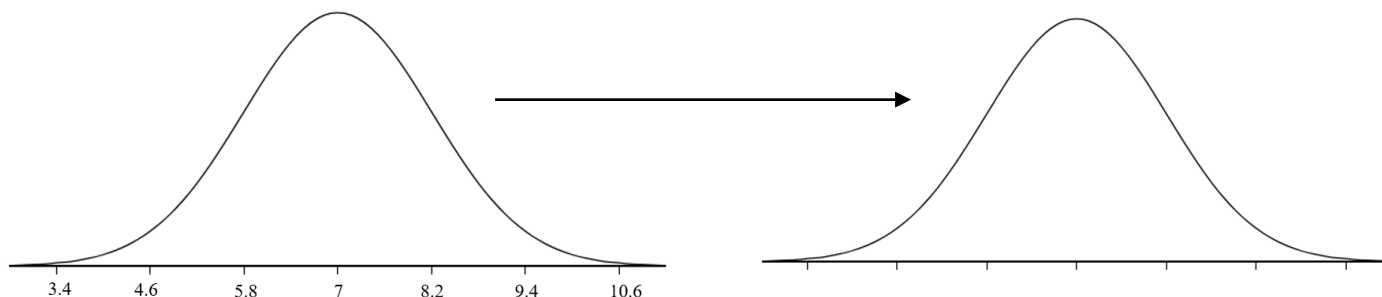


The Standard Normal Curve

How do we determine percentages if they do not follow the Empirical Rule? We standardized the normal curve.



The curve we created is called the Standard Normal curve and any data set that is normally distributed can be standardized into the Standard Normal curve.

Great, why? Well, if we have a standard curve, we have a standard set of measurements we can use to determine what percentage of data falls above, below, or between specific z-scores.

The Standard Normal Probabilities Table

For any z-score, we can find the total area (percentage) of data below a z-score. If you want to know the percentage above a z-score, you must take (100% - percentage below).

Because all Normal distributions are the same when we standardize, we can find the areas under any Normal curve from a single table or using our calculator.

Standard Normal Probability Table

This is a table of areas under the standard Normal curve. The table entry for each value z is the **area under the curve to the left** of z .

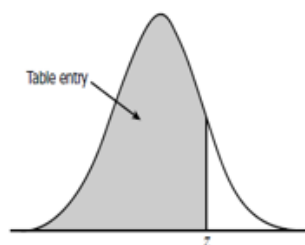


Table entry for z is the area under the standard normal curve to the left of z .

1) What percent of data is below a z-score of 0.54?

Z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9031	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9924	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9958	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986

2) What percent of data is below a z-score of 1.08?

3) What percent of data is above a z-score of 0.17?

*One side of the table is positive z-scores and the other side is negative z-scores.

The table is great, but we have a much more efficient way to get our percentages – with the calculator.

TI-84 Calculator Commands

2nd → VARS → 2: normalcdf()

LOWER: z-score

UPPER: z-score

μ : mean of z-scores → _____

σ : standard deviation of z-scores → _____

1) What percent of data is below a z-score of 0.54?

2) What percent of data is above a z-score of 0.17?

3) What percent of data lies
between $z = -1.64$ and $z = 2.33$?

Raw Data to Percentages

Example: On a typical Saturday, Fiesta Market reports that the mean amount of money spent by customers is \$27.21 and a standard deviation of \$7.93. The distribution is approximately normal.

1) What percentage of people spend less than \$20?

2) What percentage of people spend above \$40?

3) How many people spend between \$25 and \$30?

Find the percentage:

<i>Lower/less than/below</i> What percent of students scored less than a score of x ?	$z_{\#} = \frac{x - \mu}{\sigma}$ $P(z \leq z_{\#})$ *Unbounded to the left	Lower: -1e99 Upper: $z_{\#}$ μ : 0 σ : 1
<i>Greater/more than/above</i> What percent of students scored above a score of x ?	$z_{\#} = \frac{x - \mu}{\sigma}$ $P(z \geq z_{\#})$ *Unbounded to the right	Lower: $z_{\#}$ Upper: 1e99 μ : 0 σ : 1

<i>Between</i> What percent of students scored between x_{low} and x_{high}?	$z_{low} = \frac{x_{low} - \mu}{\sigma}$ $z_{high} = \frac{x_{high} - \mu}{\sigma}$ $P(z_{low} \leq z \leq z_{high})$ *Unbounded to the left	Lower: z_{low} Upper: z_{high} μ: 0 σ: 1
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Percentages to Raw Data

This time, we take a percentage below, above, or between, and we work “backwards” to find the data value that it corresponds to in the distribution.

Example Question: On a typical school night, students get an average of 7 hours of sleep with a standard deviation of 1.2 hours. How many hours of sleep would a student need to get to be in the 75th percentile?

Step 1: Find the z-score

75th percentile = 75% below Is this going to be a positive or negative z-score? How do you know?	2nd → VARS → 3:invNorm area: percentage to the left (below) as a decimal μ: 0 σ: 1 This returns the z-score that corresponds to this percentile. *The calculator ONLY does area below, so area above or between will need some problem-solving skills.
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Step 2: Find the raw data

Take your z-score, and the mean & standard deviation of your distribution, and use the z-score formula to solve for x, your raw data value.

Examples: z-scores to raw data

1) If $z = -2.3$, $\mu = 23$, and $\sigma = 5$, what is your “x” value?

Examples: percentages to z-scores

3) If you have 20% of your data lying below a z-score, what is that z-score?

4) If you have 34.5% of your data lying above a z-score, what is that z-score?

- 5) If you have 35% of the data lying symmetrically in the middle of your distribution, what are your z-scores?
- 6) What z-score corresponds to the 45th percentile?

Examples: Percentages to Raw Data

7) On the most recent test, a student scored in the 40th percentile. The mean of the test scores was an 85 and the standard deviation was 1.5. What was the student's score?

8) An automobile dealer finds that the average price of a previously owned vehicle is \$8,256. He decides to sell cars that will appeal to the middle 60% of the market in terms of price. Find the maximum and minimum prices of the cars the dealer will sell. The standard deviation is \$1,150 and the variable is normally distributed.